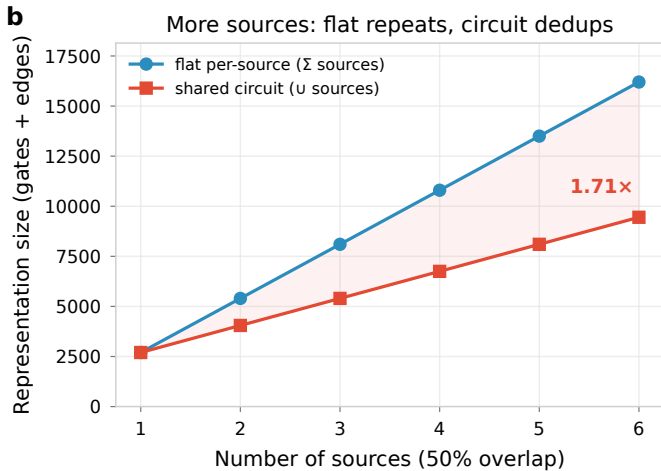
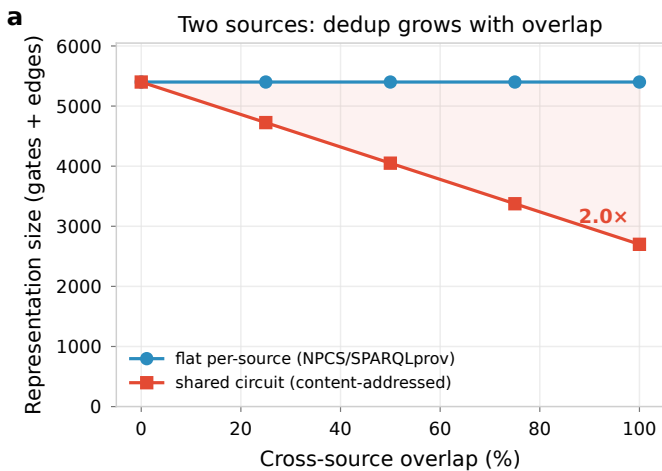




Content-addressing stores each derivation once across sources (circuit $\sim u$; flat $\sim \Sigma$) — a representation win orthogonal to construction time; it also compiles once, giving 2.9x faster end-to-end PQE at full overlap (see r9.5).